Interactions of Multiple Large Models in Representation Space

Junping Li, Yulin Bai, Shushang Wen, Jinyang Ding
Shanghai Jiao Tong University, University of Science and Technology of China

Abstract

The rapid advancement of large-scale pre-trained models across language, vision, audio, and other modalities has fundamentally transformed artificial intelligence research. While individual models have achieved remarkable performance, the interactions between multiple large models — particularly within their latent representation spaces — remain an underexplored yet critically important frontier. This survey provides a systematic review of how multiple large models interact through the representation or abstract spaces. We focus on three central paradigms: cross-modal alignment that conducts mappings of different modal spaces; homogeneous model fusion that operates within architectural manifolds; and multi-agent interaction that occurs in representation spaces or with social factors. We identify key challenges including the lack of a unified theoretical framework, limited manipulability of spaces, and insufficient cross-disciplinary integration. This work synthesizes insights from representation learning, multimodal, model merging, and multi-agent systems to establish a foundation for future research on representation or abstract space interaction.

1 Introduction

The past decade has witnessed a paradigm shift in artificial intelligence, driven by the emergence of large-scale pre-trained models that learn rich representations from massive datasets. From early distributional word representations to Transformer-based architectures (Vaswani et al., 2017) that enabled scaling to unprecedented sizes, modern AI systems now encode knowledge into high-dimensional latent spaces. Large language models (LLMs) such as GPT (Radford et al., 2018, 2019), BERT (Devlin et al., 2019), GPT-3 (Brown et al., 2020), and LLaMA (Touvron et al., 2023a), together with their successors, have demonstrated that scaling model size and training data yields emergent capabilities across diverse tasks. Parallel developments in computer vision have produced Vision Transformers (ViT) (Dosovitskiy et al., 2020), while speech research has yielded wav2vec 2.0 (Baevski et al., 2020) and Audio Spectrogram Transformers (AST) (Gong et al., 2021) that learn powerful audio representations. These models are increasingly unified by fine-tuning and reinforcement learning from human feedback (Ouyang et al., 2022; Schulman et al., 2017) and by retrieval augmentation (Lewis et al., 2020), while the capabilities and risks of such foundation models have been systematically catalogued (Bommasani et al., 2021).

The study of these specific models creates an ecosystem with different representation spaces. Each model maps its input mode to a high-dimensional vector space, in which semantic information is encoded as geometric structure. However, with the increase of the number and diversity of these models, a basic problem has arisen: whether and how multiple large-scale models can interact with each other through representation space, and what new results and phenomena can be generated from these interactions. Although most researches focus on the model structure and data level, from the perspective of representation space, the interaction between models, whether it is cross modal or multi-agent, is still difficult to understand.

Representation space interactions occur whenever information flows between models through their latent embeddings rather than through natural language or symbolic interfaces. Cross-modal models such as CLIP (Radford et al., 2021) and SigLIP (Zhai et al., 2023) align image and text representations into a shared space where similarity can be measured geometrically. Models like Flamingo (Alayrac et al., 2022), BLIP-2 (Li et al., 2023b), LLaVA (Liu et al., 2023), and ImageBind (Girdhar et al., 2023) extend this alignment to enable multi-modal understanding and generation by projecting representations across modality-specific spaces. Beyond cross-modal alignment, model merging techniques such as Model Soup (Wortsman et al., 2022), task arithmetic (Ilharco et al., 2022), and representation fusion methods (Yang et al., 2023) demonstrate that models can be combined directly in weight or representation space to produce composite capabilities. Most recently, multi-agent systems based on LLMs (Park et al.,

2023; Hong et al., 2024; Wu et al., 2023) have shown that societies of models interacting through natural language—and potentially through deeper representational channels—exhibit emergent social behaviors including cooperation, competition, and cultural transmission.

Despite these exciting developments, current research largely treats these paradigms in isolation. Cross-modal alignment, model fusion, and multi-agent interaction are typically studied as separate subfields with their own methodologies and terminologies. We argue that these are all instances of a more general concept: interaction between large models in representation space.

This survey makes the following contributions: (1) We provide a conceptual framework for understanding interactions of multiple large models in the perspectives of representation or abstract spaces. (2) We comprehensively review three major paradigms: cross-modal space, homogeneous model fusion, multi-agent interactions. (3) We identify critical open challenges including the need for formal theoretical frameworks, improved manipulability of spaces, and cross-disciplinary integration with cognitive science, economics, and sociology.

The remainder of this paper is organized as follows. Section 2 presents cross-modal and multi-modal space interactions. Section 3 reviews model fusion from a representation space perspective. Section 4 discusses multi-agent space interactions and social factors. Section 5 outlines future directions, and Section 6 concludes.

2 Representation Spaces and Cross-Modal Interaction

The foundation of modern representation learning is that high-dimensional data can be mapped into continuous vector spaces where geometric relationships encode semantic structure. This section traces modal representation spaces and discusses how they interact through alignment and fusion.

2.1 Modal Representation Spaces

Each perceptual modality gives rise to its own representation space, shaped by architecture and training objective. In natural language processing, Transformers (Vaswani et al., 2017) revolutionized representation learning through self-attention, enabling models to capture long-range dependencies and context-sensitive structure at scale. BERT (Devlin et al., 2019) applied masked language modeling to learn deep bidirectional representations across eleven NLP tasks. GPT-3 (Brown et al., 2020) demonstrated that autoregressive Transformers with 175B parameters exhibit few-shot capabilities, while the LLaMA series (Touvron et al., 2023a,b) catalyzed the open LLM ecosystem. These models map tokens into sequences of vectors residing in a linguistic space that encodes syntax, semantics, and world knowledge.

In vision and speech, ViT (Dosovitskiy et al., 2020) later adapted Transformers to images by splitting them into patches, yielding a spatial-aware visual representation space. Large-scale promptable models such as SAM (Kirillov et al., 2023) further show that a single visual encoder can support diverse downstream tasks. wav2vec 2.0 (Baevski et al., 2020) applies contrastive self-supervised learning directly to raw waveforms, learning discrete speech units that dramatically improve ASR with limited labels, while AST (Gong et al., 2021) extends Transformers to audio spectrograms.

Collectively, these encoders produce powerful but geometrically incompatible representations: each resides in a separate vector space with its own dimensionality, metric, and statistical regularities, and there is no natural cross-dimensional correspondence between them. The dimensionality, parameterization, and training signal of each space are chosen independently, so the resulting geometries are not merely unrelated but often incompatible: distances, directions, and neighborhoods carry different meanings in each.

2.2 Cross-Modal Alignment and Fusion

Cross-modal alignment enables interaction between these disparate spaces. CLIP (Radford et al., 2021) trains a dual-encoder projecting images and text into a shared space where cosine similarity reflects semantic correspondence, trained on 400 million pairs with contrastive loss, yielding remarkable zero-shot transfer. SigLIP (Zhai et al., 2023) replaces the softmax contrastive loss with a sigmoid loss, improving performance at smaller batch sizes. ALIGN (Jia et al., 2021) scales to over one billion noisy pairs, showing that large-scale contrastive learning aligns modalities even with minimal filtering.

More complex mechanisms enable deep multi-modal reasoning. Flamingo (Alayrac et al., 2022) introduces Perceiver Resampler modules that compress vi-

sual features into a fixed number of tokens interleaved with text, enabling few-shot multi-modal understanding while freezing pretrained components. BLIP-2 (Li et al., 2023b) proposes Q-Former, bridging modality gaps in two stages: first learning visual-language representations with frozen encoders, then connecting to a frozen LLM for generation, achieving parameter-efficient training. LLaVA (Liu et al., 2023) uses an MLP to project CLIP-ViT features into LLaMA's word embedding space, showing that a simple projection combined with instruction tuning yields strong multi-modal conversation. These designs illustrate a spectrum of alignment interfaces—contrastive objectives, learned query bottlenecks, and direct linear projection—each defining a different geometry through which two spaces exchange information.

ImageBind (Girdhar et al., 2023) learns a single joint embedding across six modalities (images, text, audio, depth, thermal, IMU) using images as an anchor, demonstrating that modalities never directly paired can still align, and enabling emergent cross-modal retrieval. Data2vec (Baevski et al., 2022) pursues unification differently, through self-supervised prediction of masked representations across speech, vision, and text. Recent surveys catalogue the rapid growth of this multi-modal LLM landscape (Yin et al., 2023; Su et al., 2023). A recurring question is whether such pairwise alignments compose transitively—if image aligns with text and text with audio, does image align with audio—and current evidence suggests that transitive alignment is approximate at best.

2.3 Challenges and Frontiers

Current cross-modal spaces face fundamental limitations. Most alignment relies on linear projections or attention bridges that may not capture the full geometric complexity of the underlying manifolds, and modalities differ in statistical properties and abstraction levels. Vision-Language-Action models such as RT-2 (Brohan et al., 2023) require alignment across perception, language, and motor control. World models (Ha and Schmidhuber, 2018) learn internal environment models for prediction and planning, but integrating them with LLMs requires spaces that encode both symbolic and spatial information. Modality gaps—where certain concepts are better represented in one modality—cause degraded reasoning and hallucination. Bridging these gaps while preserving each modality's representational strengths, and moving toward a genuinely universal space in which any model can read from and write to the representations of any other, remains a central challenge.

3 Spaces of Homogeneous Model Fusion

Beyond cross-modal interaction, another important form occurs between models of the same modality. Model fusion explores how models trained on different tasks or data can be combined for composite capabilities. Viewed representationally, fusion operates on the geometry of weight spaces and the feature manifolds they induce.

3.1 Weight Space Geometry and Model Merging

A key insight is that models initialized from the same pretrained checkpoint and fine-tuned on different tasks reside in connected weight regions where linear interpolation yields meaningful results. Model Soup (Wortsman et al., 2022) shows that averaging the weights of multiple fine-tuned models (with different hyperparameters) can outperform the best individual model at single-model inference cost. Built on Stochastic Weight Averaging (Izmailov et al., 2018), this suggests that fine-tuning explores a loss basin in which weight averaging remains high-performing, so the "souped" model captures a consensus across optimization trajectories.

Task arithmetic (Ilharco et al., 2022) reveals that task-specific knowledge corresponds to directions in weight space. If θ_{pre} is the pretrained model and θ_{task} the fine-tuned one, the task vector $\tau = \theta_{\text{task}} - \theta_{\text{pre}}$ encodes the direction of improvement. These vectors can be composed: adding transfers capabilities, negating removes behaviors, and combining several yields multi-task models without further training. This treats models as points in a vector space where arithmetic produces meaningful combinations, much as words compose in an embedding space.

However, linear interpolation fails when models follow divergent trajectories, because internal representations become permuted and rotated. Git Re-Basin (Ainsworth et al., 2022) finds permutation symmetries that align different models to a common basis, enabling merging even across independently trained models. Panariello et al. (Panariello et al., 2025) project LoRA adaptations into a "core space" for efficient merging, and TIES-Merging (Yadav et al., 2023) ad-

dresses interference by trimming redundant parameters, resolving sign conflicts, and merging disjoint parameters. Together these results show that careful handling of weight-space interference is crucial, and that alignment of coordinate frames is often a prerequisite for any meaningful interaction. Quantifying the similarity between two coordinate frames, and choosing an alignment that minimizes functional rather than merely parameter-space distance, remains an open problem.

3.2 Dimension-Level Fusion and Attention

Beyond weight interpolation, dimension-level fusion combines intermediate activations through learned projection and attention. Traditional ensembles (Breiman, 1996; Freund and Schapire, 1997; Lakshminarayanan et al., 2017) combine predictions through voting, operating purely at the output level. Deep fusion methods instead interact at intermediate layers via concatenation with projection, weighted gating, or cross-attention.

For LLMs, FuseLLM (Wan et al., 2024) and representation fusion (Yang et al., 2023) demonstrate that knowledge from multiple LLMs can be fused by aligning their spaces through linear transformations and combining at the token level, revealing that different LLMs develop comparable semantic structures that projections can align. Mixture-of-Experts models (Shazeer et al., 2017; Lepikhin et al., 2020; Fei et al., 2022) represent learned fusion, where gating networks route inputs to specialized experts and create sparse combinations of representation spaces. Adapter methods (Hu et al., 2021; Pfeiffer et al., 2021) allow lightweight specialized modules to be composed atop frozen base models, offering parameter-efficient fusion in which each adapter encodes a small, reusable displacement within the shared space.

Cross-attention serves as the primary interface where spaces interact: one model's representations act as keys and values while another provides queries, enabling bidirectional information flow. Attention is thus a differentiable mechanism for dynamically selecting and weighting information from different sources, providing a flexible channel for dimension-level interaction that subsumes concatenation and gating as special cases. Because attention weights are input-dependent, the resulting combination is not a fixed point in a single space but a trajectory through a family of spaces, which helps explain both the flexibility and the unpredictability of learned fusion.

3.3 Discussion

Despite progress, fundamental challenges remain. The abstract nature of high-dimensional spaces makes it difficult to predict how merging affects specific capabilities, and weight interpolation is sensitive to loss landscape connectivity, so models in separate basins produce degraded performance. Alignment between heterogeneous architectures remains poorly understood, particularly when models differ in depth or width. Most current fusion methods operate at the implementation level, lacking principled interfaces for flexible composition. Developing techniques for navigating, composing, and transforming representation spaces in a controllable manner - analogous to linear algebra for vector spaces - is therefore critical. Understanding the geometry and topology of these spaces, including task-relevant subspaces and geodesics between capabilities, will advance model fusion from heuristic practice to principled engineering. Progress here would also make fusion auditable: a principled account of which subspaces carry which capabilities would let practitioners compose and unlearn behaviors deliberately rather than by trial and error.

4 Multi-Agent Space Interaction

The most complex interactions between large models occur in multi-agent systems, where LLM-based agents interact over time and produce collective behaviors that no single model exhibits. These interactions unfold not only through explicit language but also within implicit abstract spaces—social, psychological, economic, and cultural—that arise from the interaction dynamics themselves. In representation-space terms, a multi-agent society is a coupled dynamical system in which each agent's latent state is repeatedly perturbed by the outputs of its peers.

4.1 Mechanisms of Multi-Agent Interaction

Generative Agents (Park et al., 2023) demonstrated that LLM agents in a simulated town exhibit believable individual and social behaviors, including information diffusion, relationship formation, and coordinated planning. Each agent maintains a memory stream retrieved by recency, importance, and relevance, producing a social world that evolves over time.

Subsequent work explored diverse interaction topologies. CAMEL (Li et al., 2023a) proposes role-

playing with structured communicative protocols. ChatDev (Qian et al., 2024) models software development as collaboration among CEO, programmer, and reviewer roles. MetaGPT (Hong et al., 2024) encodes standard operating procedures into workflows that require structured outputs such as design documents, code, and tests. AutoGen (Wu et al., 2023) provides a flexible framework where LLM, human, and tool agents interact in customizable topologies, supporting dynamic group chat and human-in-the-loop collaboration. These systems show that coordination quality depends less on model capacity than on the topology and protocol of state exchange. This builds on cooperative multi-agent reinforcement learning (Lin et al., 2022) and cooperative AI (Dafoe et al., 2021).

4.2 Interaction of Agents in Representation Space

Most systems communicate through text, which acts as a symbolic interface between agent states. Beneath this linguistic exchange lies a deeper representational interaction: each agent's outputs arise from internal representations—beliefs, goals, and emotions—that are decoded into language, transmitted, and re-encoded into another agent's representations through comprehension. This constitutes a noisy channel between internal representation spaces, and its bottlenecks and distortions, such as misunderstanding and ambiguity, fundamentally shape multi-agent dynamics.

Richer representation-space interaction is also being explored. Agent environments allow interaction through actions that modify a shared world state, so that observing others' behavior conveys information without explicit communication. In economic simulations, agents interact through markets where prices and transactions convey preferences and beliefs. Emergent communication research (Lazaridou et al., 2016; Havrylov and Titov, 2017) shows that reinforcement-learning agents can develop compact "languages"—protocols tailored to cooperative tasks—enabling efficient coordination beyond human language. Such protocols occupy their own representation spaces, shaped by interaction pressures rather than human supervision.

Recent work pushes this idea into the models' own latent space. Cache-to-Cache (Fu et al., 2026) uses the key-value cache as the communication medium, projecting and fusing the source model's cache with that of the target model, and reports higher accuracy with roughly $2.5\times$ lower latency than text-based exchange. Interlat (Du et al., 2026) transmits continuous last hidden states so that agents communicate entirely in latent space, and shows that such messages can be compressed to a few tokens while remaining effective even across heterogeneous models. These results suggest that the noisy textual channel can be bypassed, letting agents exchange richer semantics directly within a shared or aligned latent space.

4.3 Social Interaction

Multi-agent LLM systems have become platforms for studying social phenomena. Research reveals that LLM agents exhibit behaviors consistent with game theory while displaying human-like biases and reciprocity norms (Akata et al., 2023). Economic simulations reproduce price formation, supply-demand dynamics, and market bubbles as agents trade and learn.

Collectives develop emergent cultural phenomena. Information diffusion demonstrates how beliefs spread through networks, with agents exhibiting polarization, confirmation bias, and herd behavior (Gao et al., 2023). Artificial Leviathan experiments (Chen, 2024) show that agents in resource-scarce environments initially exhibit Hobbesian conflict, then progressively develop social contracts and cooperative institutions. CiteAgent (Liu et al., 2025) simulates citation networks that reproduce the power-law distributions and citational distortion characteristic of academia. Emotion and affect also emerge, influencing group dynamics, cooperation, and decision-making. These phenomena reside in the interactions between agents rather than within any single model, constituting emergent social spaces invisible to any individual parameter set.

4.4 Social Science Disciplines and Social Factors

More social science disciplines or social factors can affect multi-agent interaction. Psychology provides insights into how memory, attention, and reasoning affect interaction. Theory of Mind - attributing mental states to others - is crucial for collaboration (Sap et al., 2022; Shapira et al., 2024). Behavioral economics provides frameworks for bounded rationality and social preferences, while sociological theories of networks, institutions, and cultural evolution inform agent society design. However, a fundamental challenge remains: unlike human interactions grounded in physical reality, LLM agent societies are grounded only in text and de-

signed environments. Understanding how this difference shapes social spaces of agents, and how it contributes to human society, requires the combination of agents and social science.

5 Problems and Future Directions

Despite significant progress across cross-modal alignment, model fusion, and multi-agent interaction, several fundamental challenges must be addressed.

Theoretical Framework. A primary limitation is the absence of formal theory for representation space interaction. Existing work is largely empirical: contrastive losses for alignment, weighted averaging for merging, prompt engineering for communication. What is needed is a mathematical theory characterizing when spaces can be aligned, what geometric operations correspond to compositional capabilities, and how information flows between models. Such a framework would draw on differential geometry (modeling representation manifolds), information theory (quantifying inter-model channels), and category theory (formalizing structure-preserving mappings). A critical open question is whether universal representations exist through which arbitrary models can interact, analogous to natural language for humans but optimized for machine-to-machine transfer.

Manipulability and Composability. Current approaches remain at low-level code manipulation rather than principled interfaces. Researchers merging models, aligning modalities, or connecting agents implement custom pipelines with limited composability. There is need for standardized abstractions treating representation spaces as first-class objects, supporting projection, interpolation, composition, and decomposition into interpretable subspaces, enabling flexible combination and editing of model capabilities. Such a system would dramatically lower barriers to building multi-model systems and sophisticated models.

Cross-Disciplinary Integration. Representation space interaction can be combined with the theories of other disciplines. For example, multi-agent can combine economics, sociology and psychology to understand or solve the phenomena or problems of human society. Yet current AI research often develops in isolation. Deeper engagement with cognitive science, social network theory, game theory, linguistics, and theory of mind could provide algorithmic inspiration and behavioral validation. Understanding how human minds, groups and societies develop and evolve norms provides rich hypotheses for designing multi-model systems that work more effectively.

6 Conclusion

This survey is mainly about interactions of multiple large models through the perspectives of representation space. We reviewed three central paradigms: cross-modal alignment that bridges vision, language, audio, and other modalities into shared geometric spaces through contrastive learning and projection mechanisms; model fusion that composes capabilities within weight and representation spaces through averaging, arithmetic, and attention-based combination; and multi-agent interactions in representation spaces or with social factors such as cooperation, competition, culture, and economics.

Across these paradigms, representation space interaction reveals a fundamental shift in perspective: from viewing AI systems as individual objects to viewing them as components within larger ecosystems of interacting representational structures. The unified theoretical framework, representation space operations, and cross-disciplinary integration will be essential to advance this field. As large models become increasingly central to AI systems, understanding how they interact, combine, and collectively produce intelligence suggests new avenues for research.

References

Samuel K Ainsworth, Jonathan Hayase, and Siddhartha Srinivasa. 2022. Git re-basin: Merging models modulo permutation symmetries. In *International Conference on Learning Representations*.

Emre Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2023. Playing repeated games with large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 11098–11114.

Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc,

Arthur Mensch, Katie Millican, Malcolm Reynolds, et al. 2022. Flamingo: A visual language model for few-shot learning. In *Advances in Neural Information Processing Systems*, volume 35, pages 23716–23736.

Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jeff Gu, and Michael Auli. 2022. data2vec: A general framework for self-supervised learning in speech, vision and language. In *International Conference on Machine Learning*, pages 1298–1312.

Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems*, volume 33, pages 12449–12460.

Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. 2021. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*.

Leo Breiman. 1996. Bagging predictors. *Machine Learning*, 24(2):123–140.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. 2023. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901.

Peng Chen. 2024. Artificial leviathan: Exploring social evolution of llm agents through the lens of hobbesian social contract theory. *arXiv preprint arXiv:2406.14373*.

Allan Dafoe, Yoram Bachrach, Gillian Hadfield, Eric Horvitz, Kate Larson, and Thore Graepel. 2021. Open problems in cooperative ai. *arXiv preprint arXiv:2012.08630*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, volume 1, pages 4171–4186.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*.

Zhuoyun Du, Runze Wang, Huiyu Bai, Zouying Cao, Xiaoyong Zhu, Yu Cheng, Bo Zheng, Wei Chen, and Haochao Ying. 2026. Enabling agents to communicate entirely in latent space. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 27106–27129.

Tianyi Fei, Tao Shen, Yue Yu, Xian Wu, Yiming Chen, Tianlong Chen, Zhen Li, and Weizhu Chen. 2022. Mixture-of-experts meets instruction tuning: A winning combination for large language models. *arXiv preprint arXiv:2205.01068*.

Yoav Freund and Robert E Schapire. 1997. A decision-theoretic generalization of on-line learning and an application to boosting. In *European Conference on Computational Learning Theory*, pages 23–37.

Tianyu Fu, Zihan Min, Hanling Zhang, Jichao Yan, Guohao Dai, Wanli Ouyang, and Yu Wang. 2026. Cache-to-cache: Direct semantic communication between large language models. In *International Conference on Learning Representations*.

Chen Gao, Xiaochong Lan, Zhihong Lu, Jinzhu Mao, Jinghua Piao, Huandong Wang, Depeng Jin, and Yong Li. 2023. S3: Social-network simulation system with large language model-empowered agents. *arXiv preprint arXiv:2307.05804*.

Rohit Girdhar, Alaaeldin El-Nouby, Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. 2023. Imagebind: One embedding space to bind them all. In *Proceedings of the*

IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 15180–15190.

Yuan Gong, Yu-An Chung, and James Glass. 2021. Ast: Audio spectrogram transformer. In *Interspeech*, pages 571–575.

David Ha and Jürgen Schmidhuber. 2018. World models. *arXiv preprint arXiv:1803.10122*.

Serhii Havrylov and Ivan Titov. 2017. Emergence of language with multi-agent games: Learning to communicate with sequences of symbols. In *Advances in Neural Information Processing Systems*, volume 30.

Sirui Hong, Mingchen Zhuge, Jiaqi Chen, Xiawu Wu, Shenghao Cheng, Junnan Liu, Chenghao Zhang, Yu Wang, et al. 2024. Metagpt: Meta programming for a multi-agent collaborative framework. In *International Conference on Learning Representations*.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Gabriel Ilharco, Tulsee Tulabandhula, Mitchell Wortsman, Carlos Guestrin, Ali Farhadi, and Hannaneh Hajishirzi. 2022. Editing models with task arithmetic. In *International Conference on Learning Representations*.

Pavel Izmailov, Dmitrii Podoprikhin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. 2018. Averaging weights leads to wider optima and better generalization. In *Conference on Uncertainty in Artificial Intelligence*, pages 876–885.

Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V Le, Yunhsuan Sung, Zhen Li, and Tom Duerig. 2021. Scaling up visual and vision-language representation learning with noisy text supervision. In *International Conference on Machine Learning*, pages 4904–4916.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. 2017. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30.

Angeliki Lazaridou, Alexander Peysakhovich, and Marco Baroni. 2016. Multi-agent cooperation and the emergence of (natural) language. In *International Conference on Learning Representations*.

Dmitry Lepikhin, HyoukJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. 2020. Gshard: Scaling giant models with conditional computation and automatic sharding. In *International Conference on Learning Representations*.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Guohao Li, Hasan Alrashed, Soham Sengupta, Ziwei Zeng, Daniel Khashabi, and Tushar Khot. 2023a. Camel: Communicative agents for "mind" exploration of large language model society. *arXiv preprint arXiv:2303.17760*.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023b. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning*, pages 19730–19742.

Kai Lin, Renhe Zhao, Zhe Xu, and Jiayu Zhou. 2022. A survey of multi-agent reinforcement learning: Methods and applications. In *International Conference on Machine Learning*.

Feng Liu et al. 2025. Leveraging llm-based agents for social science research: insights from citation network simulations. *Humanities and Social Sciences Communications*, 12:1–14.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual instruction tuning. In *Advances in Neural Information Processing Systems*, volume 36.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.

Aniello Panariello et al. 2025. Accurate and efficient low-rank model merging in core space. In *Advances in Neural Information Processing Systems*.

Joon Sung Park, Joseph C O'Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22.

Jonas Pfeiffer, Aishwarya Kamath, Andreas Rücklé, Kyunghyun Cho, and Iryna Gurevych. 2021. Adapterfusion: Non-destructive task composition for transfer learning. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 487–503.

Chen Qian, Wei Cong, Yang Liu, Weize Chen, Yang Wang, Xin Su, Yufan Dang, Jiahao Li, Jiahao Xu, Dahai Li, et al. 2024. Communicative agents for software development. In *International Conference on Software Engineering*, pages 1577–1589.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763.

Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. 2018. Improving language understanding by generative pre-training. In *OpenAI Technical Report*.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners. *OpenAI Blog*, 1(8).

Maarten Sap, Ronan Le Bras, Daniel Fried, and Yejin Choi. 2022. Neural theory-of-mind? on the limits of social intelligence in large lms. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 7362–7381.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.

Natalie Shapira, Mosh Levy, Seyed Hossein Alavi, et al. 2024. Clever hans or neural theory of mind? stress testing social reasoning in large language models. In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 1912–1933.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*.

Yifan Su, Tian Lan, Huayang Li, Yang Liu, Jun Xu, Wenwu Wang, and Deng Cai. 2023. A comprehensive evaluation of multi-modal large language models. *arXiv preprint arXiv:2306.13394*.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023a. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023b. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30.

Tao Wan, Wenqian Lu, Chengwei Ding, Ruochen Xu, Jun Wang, Beidou Chen, Enhong Yang, and Yanyan Zhao. 2024. Fusellm: Knowledge fusion of large language models. In *Proceedings of the 62nd Annual*

Meeting of the Association for Computational Linguistics, pages 1608–1622.

Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, et al. 2022. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, pages 23965–23998.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Chi Gao, et al. 2023. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*.

Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. 2023. Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, volume 36.

Junfeng Yang, Jing Liu, Xin Wang, Hongxia Zhao, and Yun-Nung Chen. 2023. Fusing models via representation space alignments. In *Advances in Neural Information Processing Systems*, volume 36.

Shukang Yin, Chaoyou Fu, Sirui Zhao, Ke Li, Xing Sun, Tong Xu, and Enhong Chen. 2023. A survey on multimodal large language models. *arXiv preprint arXiv:2306.13549*.

Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11975–11986.